

D603

Machine Learning

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A fraud classifier at a card issuer that processes 2 million transactions a day flagged 200 transactions, and 150 of them were truly fraud. There were 250 fraudulent transactions in total. The model was built in scikit-learn. What is the model's precision?

- A. 0.60
- B. 0.25
- C. 0.80
- D. 0.75

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A model has 90 true positives and a recall of 0.60. How many actual positive cases were there?

- A. 60
- B. 90
- C. 150
- D. 54

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A rare-disease screening model at a 40-lab testing network reports 99% accuracy on data in which 1% of patients have the disease. The network recently won a state contract. What is the main concern?

- A. The new contract means more data is needed first.
- B. Predicting “no disease” for everyone would also score 99%; check recall and precision.
- C. None; 99% accuracy proves the model is excellent.
- D. The accuracy is too low for medical use.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A kNN model at a real-estate site uses square footage (500 to 5,000) and bathroom count (1 to 4) to find similar homes. The site gets 300,000 visits a month. Predictions seem to ignore bathrooms. Why?

- A. k is too large.
- B. Site traffic is too high for kNN.
- C. kNN cannot use whole-number features.
- D. Square footage dominates the distance; scale both features first.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A kNN model with $k = 1$ has near-perfect training accuracy but poor test accuracy. Which change is most likely to help?

- A. Set k to the full training size, which is always most stable.
- B. Increase k and pick it by cross-validation.
- C. Stop scaling the features.
- D. Lower k further.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A churn decision tree at a wireless carrier, grown with no depth limit on 80,000 customers, has 4,000 leaves, 100% training accuracy, and 61% test accuracy. The carrier's stores close at 9 p.m. What is the best fix?

- A. Limit depth or minimum leaf size, or prune the tree.
- B. Let the tree grow deeper.
- C. Add store hours as a feature.
- D. Switch the split criterion from Gini to entropy.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

An SVM separating legitimate emails from spam at a 900-employee firm finds the two classes are not linearly separable in the original features. The IT team tracks work in a ticketing tool. Which approach fits?

- A. Switch to k-means clustering.
- B. Abandon SVMs, since they only work on linearly separable data.
- C. Use a kernel (such as RBF) so the SVM can draw a non-linear boundary.
- D. Choose the boundary that passes closest to the data points.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

An analyst at a fitness app with 2.5 million users runs k-means for $k = 1$ to 10. Inertia drops sharply up to $k = 4$ and flattens after that. The app's color scheme is orange. Which k does the elbow method suggest?

- A. 1
- B. It cannot be chosen without running hierarchical clustering first
- C. 4
- D. 10, because the lowest inertia is always best

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A team wants to group customers without fixing the number of clusters in advance and wants to see how groups nest inside larger groups. Which method fits?

- A. Hierarchical clustering with a dendrogram
- B. k-nearest neighbors
- C. k-means
- D. t-SNE

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A t-SNE plot of 50,000 customer profiles at an e-commerce firm shows cluster A far from cluster B and close to cluster C. The plot was presented at an all-hands meeting. Which conclusion is justified?

- A. Points within each cluster are similar, but distances between clusters are not reliable.
- B. Use the t-SNE map to score new customers.
- C. A is more similar to C than to B.
- D. The plot proves there are exactly three segments.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A single deep decision tree at a claims processor with 17 regional offices gives very different results on different samples. The team wants many trees trained independently on bootstrap samples with random feature subsets, then averaged. Which method is this?

- A. Gradient boosting
- B. Random forest (bagging)
- C. One pruned tree
- D. k-nearest neighbors

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

Which ensemble method trains models one after another, with each new model focusing on the errors of the previous ones?

- A. Bagging
- B. Random forest
- C. Stacking
- D. Boosting

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A fraud classifier at a card issuer ~~that processes 2 million transactions a day~~ flagged 200 transactions, and 150 of them were truly fraud. There were 250 fraudulent transactions in total. ~~The model was built in scikit learn.~~ What is the model's precision?

Target word: precision **Decides:** 150 true among 200 flagged.

A	Confusion set	150 / 250 is recall.
B	Inverted operation	50 / 200 is the share of flags that were false.
C	Wrong stem number	200 / 250 divides two stem counts that do not form either metric.
D	Correct	150 / 200 = 0.75.

2. Answer C Concept 1 · transfer

A model has 90 true positives and a recall of 0.60. How many actual positive cases were there?

Target word: solve for actual positives **Decides:** Recall = TP / actual positives.

A	Adjacent category	That is the false negatives (150 – 90).
B	Wrong stem number	The true positives from the stem.
C	Correct	90 / 0.60 = 150.
D	Inverted operation	Multiplies 90 by 0.60 instead of dividing.

3. Answer B Concept 2

A rare-disease screening model ~~at a 40 lab testing network~~ reports 99% accuracy on data in which 1% of patients have the disease. ~~The network recently won a state contract.~~ What is the main concern?

Target word: what the accuracy hides **Decides:** Only 1% are positive.

A	Camouflage	The contract does not decide the metric issue.
B	Correct	Accuracy is misleading when classes are imbalanced.
C	Absolute language	The majority class alone yields that score.
D	Inverted operation	The problem is that accuracy is uninformative here, not low.

4. Answer D Concept 3

A kNN model at a real-estate site uses square footage (500 to 5,000) and bathroom count (1 to 4) to find similar homes. ~~The site gets 300,000 visits a month.~~ Predictions seem to ignore bathrooms. Why?

Target word: why one feature is ignored **Decides:** Features are on very different scales.

A	Adjacent category	k affects smoothing, not which feature dominates.
B	Camouflage	Visit count does not affect feature weighting.
C	Absolute language	Integers work fine once scaled.
D	Correct	kNN distances are driven by the feature with the largest range.

5. Answer B Concept 3 · transfer

A kNN model with $k = 1$ has near-perfect training accuracy but poor test accuracy. Which change is most likely to help?

Target word: fix overfitting in kNN **Decides:** $k = 1$ has high variance.

A	Absolute language	That predicts the majority class for everyone and underfits.
B	Correct	Averaging more neighbors smooths the boundary.
C	Dropped element	Scaling is needed for meaningful distances.
D	Inverted operation	$k = 1$ is already the most variable setting.

6. Answer A Concept 4

A churn decision tree at a wireless carrier, grown with no depth limit on 80,000 customers, has 4,000 leaves, 100% training accuracy, and 61% test accuracy. ~~The carrier's stores close at 9 p.m.~~ What is the best fix?

Target word: fix the tree **Decides:** Unlimited depth; train-test gap.

A	Correct	Constraining the tree reduces memorization.
B	Inverted operation	More depth increases overfitting.
C	Camouflage	Store hours do not address the overfitting.
D	Adjacent category	Criterion choice has little effect on overfitting.

7. Answer C Concept 5

An SVM separating legitimate emails from spam ~~at a 900-employee firm~~ finds the two classes are not linearly separable in the original features. ~~The IT team tracks work in a ticketing tool.~~ Which approach fits?

Target word: SVM for non-linear data **Decides:** Classes are not linearly separable.

A	Adjacent category	Unsupervised; ignores the labels.
B	Absolute language	Kernels and soft margins handle this case.
C	Correct	Kernels map data to a space where a separating margin exists.
D	Inverted operation	SVMs maximize the margin, not minimize it.

8. Answer C Concept 6

An analyst at a fitness app ~~with 2.5 million users~~ runs k-means for $k = 1$ to 10. Inertia drops sharply up to $k = 4$ and flattens after that. ~~The app's color scheme is orange.~~ Which k does the elbow method suggest?

Target word: k from the elbow **Decides:** The bend is at $k = 4$.

A	Wrong stem number	The starting k from the stem.
B	Adjacent category	The elbow gives a choice directly.
C	Correct	The point where adding clusters stops paying off.
D	Absolute language	Inertia always falls as k rises; the lowest value just means the most clusters.

9. Answer A Concept 6 • transfer

A team wants to group customers without fixing the number of clusters in advance and wants to see how groups nest inside larger groups. Which method fits?

Target word: method without preset k **Decides:** No preset k; nested structure.

A	Correct	Builds nested clusters; k can be chosen by cutting the tree later.
B	Confusion set	A supervised classifier despite the similar name.
C	Dropped element	Needs k up front and gives flat clusters.
D	Adjacent category	Visualizes structure but does not assign nested clusters.

10. Answer A Concept 7

A t-SNE plot of 50,000 customer profiles at an e-commerce firm shows cluster A far from cluster B and close to cluster C. ~~The plot was presented at an all-hands meeting.~~ Which conclusion is justified?

Target word: what t-SNE supports **Decides:** t-SNE keeps local neighborhoods, not global distances.

A	Correct	t-SNE distorts global distances and cluster sizes.
B	Adjacent category	t-SNE is for visualization and has no transform for new data.
C	Confusion set	Reads global distance as in PCA; t-SNE does not preserve it.
D	Absolute language	Cluster appearance depends on settings like perplexity.

11. Answer B Concept 8

A single deep decision tree at a claims processor ~~with 17 regional offices~~ gives very different results on different samples. The team wants many trees trained independently on bootstrap samples with random feature subsets, then averaged. Which method is this?

Target word: name the ensemble **Decides:** Independent trees on bootstrap samples, averaged.

A	Confusion set	Boosting builds trees one after another on prior errors.
B	Correct	Averaging independent trees reduces variance.
C	Dropped element	No ensemble and no averaging.
D	Adjacent category	Not an ensemble of trees.

12. Answer D Concept 8 • transfer

Which ensemble method trains models one after another, with each new model focusing on the errors of the previous ones?

Target word: name the sequential ensemble **Decides:** Sequential, error-focused training.

A	Inverted operation	Bagging trains models independently in parallel.
B	Confusion set	A bagging method with random feature subsets.
C	Adjacent category	Combines different models with a meta-learner rather than sequential error fixing.
D	Correct	Each learner corrects the last, mainly reducing bias.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Precision and recall	Precision = $TP / (TP + FP)$, the share of flagged cases that are real; recall = $TP / (TP + FN)$, the share of real cases that were caught.	Swapping the two denominators.	“Flagged” counts next to “actual” counts.
2	The accuracy paradox	With imbalanced classes, high accuracy can come from always predicting the majority class; use recall, precision, or F1 instead.	Treating 99% accuracy on rare-event data as proof of a good model.	A rare positive class and a very high accuracy figure.
3	k-nearest neighbors	kNN relies on distance, so features must be scaled; very small k overfits (high variance), larger k smooths.	Running kNN on raw units, or lowering k to fix overfitting.	Features in very different units, or k = 1 with a train-test gap.
4	Decision tree overfitting	Unlimited-depth trees memorize training data; limit depth or leaf size, or prune.	Growing the tree further to fix poor test accuracy.	Thousands of leaves and perfect training accuracy.
5	Support vector machines	An SVM picks the boundary with the widest margin; kernels let it separate data that is not linearly separable.	Believing SVMs only work on linearly separable data or pick the boundary closest to the points.	Classes that overlap in the original feature space.
6	Choosing clusters	k-means needs k in advance (the elbow in inertia guides it); hierarchical clustering builds nested groups without fixing k first.	Picking the k with the lowest inertia, which always favors more clusters.	An inertia curve that bends, or a wish not to fix k.
7	Reading t-SNE	t-SNE preserves local neighborhoods for visualization; distances between clusters and cluster sizes are not reliable, and it does not score new data.	Reading global distances off a t-SNE plot or using it as a model.	Conclusions about how far apart clusters are on a t-SNE plot.
8	Bagging vs. boosting	Bagging (random forest) trains models independently on bootstrap samples and averages them to cut variance; boosting trains sequentially on prior errors to cut bias.	Swapping which is parallel and which is sequential.	“Independently and averaged” vs. “each fixes the last one’s errors.”

Course scope (public description): Machine Learning comprises the broad discipline of developing algorithms and statistical models to predict, classify, or cluster data and iteratively improve over time. Machine Learning focuses on building, training, running, and testing supervised and unsupervised models and quantifying the accuracy and precision of those models to determine which may best be used in a particular business situation. Supervised methods discussed include k-nearest neighbors, decision trees, and support vector machines. Unsupervised models discussed include k-means clustering, hierarchical clustering, and t-distributed stochastic neighbor embedding (t-SNE). Ensemble methods are also presented. The following courses are prerequisites: Analytics Programming and Statistical Data Mining.